

Nmap Server Role & Security Analysis Lab - Answer Key

Server	Role	IP Address	Non-Compliant Port
Server 1	Database	192.168.1.10	53
Server 2	Switch	192.168.1.20	23
Server 3	Web Server	192.168.1.30	80
Server 4	File Server	192.168.1.40	21
Server 5	Mail Server	192.168.1.50	25

Key Security Findings:

- Port 53 (DNS) should not be exposed on a dedicated database server.
- Port 23 (Telnet) is insecure and transmits credentials in cleartext.
- Port 80 (HTTP) is unencrypted and should be disabled in favor of HTTPS.
- Port 21 (FTP) is insecure and should be replaced with secure file transfer methods.
- Port 25 (SMTP) allows unencrypted mail transfer and should be secured or restricted.